Properties of parallelograms

1

a $x=115^{\circ}$ b $x=101^{\circ}$ c $x=30$ units d $x=8$ units

e $x=8$ units

2

a $x=4, y=69$

b $a=57, b=11$

c $x=17, y=7$

d $x=4, y=137, z=43$

e $x=7, y=16$

f $x=5, y=8$

g $x=4, y=11, z=8$

h $a=89, b=101$

i $k=7, m=8$

j $p=5, q=9$

k $w=1, z=3$

l $a=12$

3 $x=2$ m

4

a $x=9, y=7$

b 20 units

5

a

i Slope of $\overline{AB}=\frac{1}{5}$, Slope of $\overline{CD}=\frac{1}{5}$

ii Slope of $\overline{AD}=-\frac{5}{3}$, Slope of $\overline{BC}=-\frac{5}{3}$

b

i No ii Yes iii No iv No

6

a No b No c Yes d No

7 If a quadrilateral is a parallelogram, then its consecutive interior angles are supplementary.

8

To prove: $\triangle$[redacted] $\cong \triangle CDA$

	Statements	Reasons
1.	$ABCD$ is a parallelogram	Given
2.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3.	$\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$	Alternate Interior Angles Theorem
4.	$\overline{AC} \cong \overline{AC}$	Reflexive Property of Congruence
5.	$\triangle ABC \cong \triangle CDA$	Angle-Side-Angle Congruence

9

To prove: Opposite sides of a parallelogram are congruent

	Statements	Reasons
1.	$ABCD$ is a parallelogram	Given
2.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3.	$\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$	Alternate Interior Angles Theorem
4.	$\overline{AC} \cong \overline{AC}$	Reflexive property of congruence
5.	$\triangle ABC \cong \triangle CDA$	Angle-Side-Angle Congruence
6.	$\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$	Corresponding parts of congruent triangles are congruent (CPCTC)

10

To prove: Consecutive angles are supplementary.

	Statements	Reasons
1.	$PQRS$ is a parallelogram	Given
2.	$\overline{PQ} \parallel \overline{SR}$ and $\overline{PS} \parallel \overline{QR}$	Definition of a parallelogram
3.	$\angle P$ and $\angle S$, $\angle S$ and $\angle R$, and $\angle R$ and $\angle Q$ are supplementary.	Consecutive interior angles postulate
4.	$\angle P \cong \angle R$ and $\angle Q \cong \angle S$	Congruent supplements theorem

11

To prove: Opposite sides o[redacted]rallelogram are congruent

	Statements	Reasons
1.	$ABCD$ is a parallelogram	Given
2.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3.	$\angle ABD \cong \angle CDB$ and $\angle ADB \cong \angle CBD$	Alternate Interior Angles Theorem
4.	$\overline{BD} \cong \overline{BD}$	Reflexive property of congruence
5.	$\triangle ABD \cong \triangle CDB$	Angle-Side-Angle congruence theorem
6.	$\overline{AB} \cong \overline{CD}$	Corresponding parts of congruent triangles are congruent (CPCTC)
7.	$\angle BAC \cong \angle DCA$	Alternate Interior Angles Theorem
8.	$\triangle ABP \cong \triangle CDP$	Angle-Side-Angle congruence theorem
9.	$\overline{BP} \cong \overline{DP}$ and $\overline{AP} \cong \overline{CP}$	Corresponding parts of congruent triangles are congruent (CPCTC)
10.	$\overline{AC}$ bisects $\overline{BD}$ and $\overline{BD}$ bisects $\overline{AC}$	Definition of a bisector

12

To prove: COnsecutive angles of a parallelogram are supplementary

	Statements	Reasons
1.	$PQRS$ is a parallelogram	Given
2.	$\overline{PQ} \parallel \overline{SR}$ and $\overline{PS} \parallel \overline{QR}$	Definition of a parallelogram
3.	$\angle P$ and $\angle S, \angle S$ and $\angle R,$ and $\angle R$ and $\angle Q$ are supplementary.	Consecutive interior angles postulate

13 The reason in line 6 uses the result of the proof, so it cannot be used.

14 Line 3 should follow after line 1.

Additional questions

15

a True

b False

c False

d True

e True

f False

16

a Yes

b No

c No

d Yes

17

a $x = 5, y = 8$

b $P = 44$ units

18

a $KF = 16$

b $m\angle H = 55^\circ$

c $m\angle H = 53^\circ$

19 $y = 65$

20 $LN = 32$ units

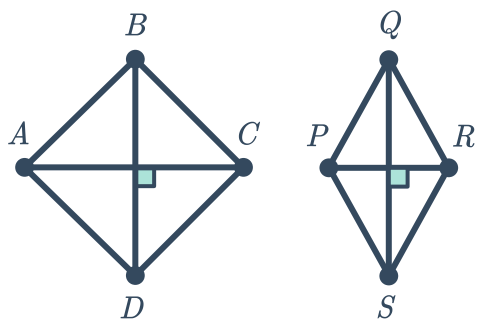
21

a $\angle SPR \cong \angle PRQ$

b $\overline{PQ} \cong \overline{SR}$

22

a



b There is not enough information to conclude that $ABCD \cong PQRS$. We would need more information about the lengths of the sides or the diagonals in order to make that conclusion.

c Any of the following pieces of information about each quadrilateral: the diagonals bisect on another, they contain two pairs of opposite sides, they each contain opposite angles that are congruent, or they each contain adjacent angles that are supplementary.

23 Example proof

To prove: $ABCD$ is a parallelogram

	Statements	Reasons
1.	$\overline{AB} \cong \overline{CD}$	Given
2.	$\angle 1 \cong \angle 2$	Given
3.	$\overline{AB} \| \overline{CD}$	Converse of the alternate interior angles theorem
4.	$ABCD$ is a parallelogram	Definition of parallelogram

24 Option 1: It is possible that the garden could be a parallelogram, but not guaranteed. We would need one more piece of information such as whether or not the diagonals bisect one another in order to conclude the garden design is a parallelogram.
Option 2: Does not form a parallelogram, since the opposite angles are not supplementary.
Option 3: It is possible that the garden could be a parallelogram, but not guaranteed. We would need one more piece of information such as whether or not the opposite sides are parallel in order to conclude the garden design is a parallelogram.

25 Example proof

To prove: E is the midpoint of $\overline{AC}$

	Statements	Reasons
1.	Parallelogram $ABCD$	Given
2.	$\overline{AB} \| \overline{DC}$	Definition of parallelogram
3.	$\overline{AB} \cong \overline{DC}$	Definition of parallelogram
4.	$\angle 2 \cong \angle 4$	Alternate interior angles theorem
5.	$\angle 1 \cong \angle 3$	Alternate interior angles theorem
6.	$\triangle AEB \cong \angle CED$	ASA congruence theorem
7.	$\overline{AE} \cong \overline{EC}$	CPCTC
8.	E is the midpoint of $\overline{AC}$	Definition of midpoint